Retrofitting Summary Document

Status as of March 2017

Introduction

Housing was the most affected sector following the April 25th 2015 Gorkha earthquake. The initial focus has been on operationalising the 300,000 NPRs reconstruction grant from the Government of Nepal (GoN). The grant disbursement guidelines were recently revised to include provision for a 100,000 NPRs grant for retrofitting, but the mechanisms for disbursing the grant are yet to fully developed and operationalised.

The revised grant disbursement guidelines and recently approved DUDBC Retrofitting Guidelines open the door for the NRA to authorise and regulate the usage of retrofitting of houses as part of the reconstruction effort. Retrofitting can help to address the following groups of homeowners within the reconstruction:

1. Houses partially damaged by the earthquake.
2. Houses rebuilt since the earthquake but not meeting the NRA/DUDBC inspection criteria.
3. Houses currently under construction but not meeting the NRA/DUDBC inspection criteria.

There are several positive outcomes gained by retrofitting – seismically strengthening and stabilising the existing structure – in preference to demolition. These include;

- A larger living space is maintained for the same cost greatly improving the value for money of reconstruction for the homeowner. The homeowner's current lifestyle and cultural conditions are also maintained: agrarian assets and lifestyle are preserved, facilitating the economic rebound of agrarian communities.
- Reduced use of materials, such as stone, rebar, gravel, timber resulting in lower environmental costs.

The roll out, and scale up, of retrofitting as part of the earthquake reconstruction, should be viewed as part of the bigger picture of Disaster Risk Reduction (DRR) efforts in Nepal. The first retrofit pilot in Nepal was implemented in 1999, and since then hundreds of buildings, especially schools and other public buildings have been retrofitted. The move to expand retrofitting at scale across Nepal is also reflected in longer term planning, such as the National Action Plan for Safer Building Construction which includes as an activity the drafting of building codes for retrofitting.

Other useful reference documents / links include the information from the retrofitting technical session held on November 22nd 2016, a GSDRC Helpdesk Research Report on the 'Seismic retrofit of earthquake damaged masonry housing', a summary of the National Society for Earthquake Technology (NSET) experience on retrofitting, and a note on 'experience in Repair and Retrofitting in the Housing Sector after the Kashmir Earthquake, Pakistan 2005'.

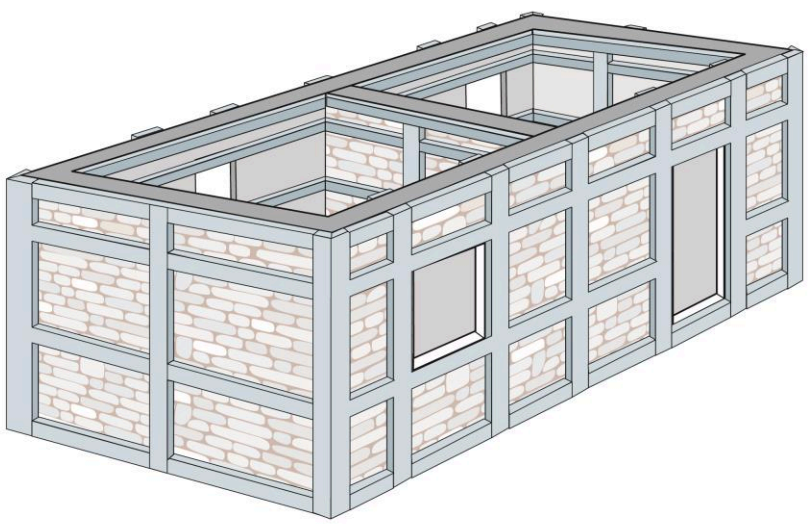
A technical working group has been established under the NRA lead by Dr. Hari Parajuli, NRA Executive Member for Housing and Settlements. The group includes technical staff from technical experts from NRA, MoUD CL-PIU, NSET, Build Change, JICA, UNDP, and HRRP. POs interested to commit technical expertise to the working group may contact Narayan Marasini, HRRP National Technical Coordinator, for further details (9808565098, techcord.national@hrrpnepal.org). This document provides a summary of some on-going retrofitting activities.

Corrections Manual

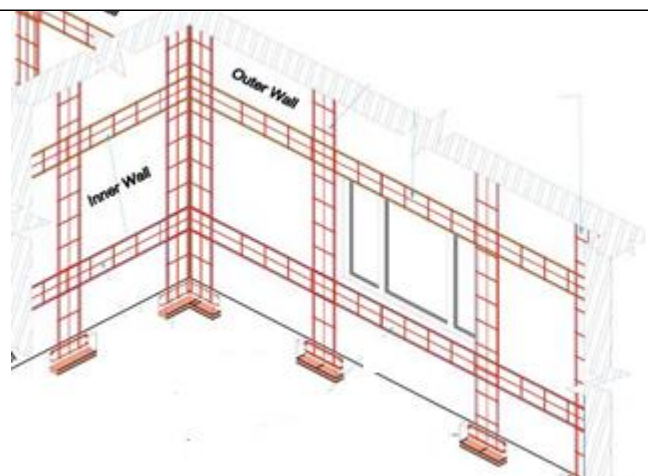
JICA have initiated preparation of a 'Corrections / Exceptions Manual' which will document non-compliances and provide guidance on remedial measures to correct these, as well as identifying exceptions where should all other Minimum Requirements be met, there may not need to be corrections carried out. The current tentative plan is to finish the initial version of the manual (it will be a living document that will be updated to include additional non-compliances as and when required) in March and then roll out orientation on the manual to DL-PIU engineers in the field.

Handbook for Repair and Retrofitting of Stone in Mud Buildings

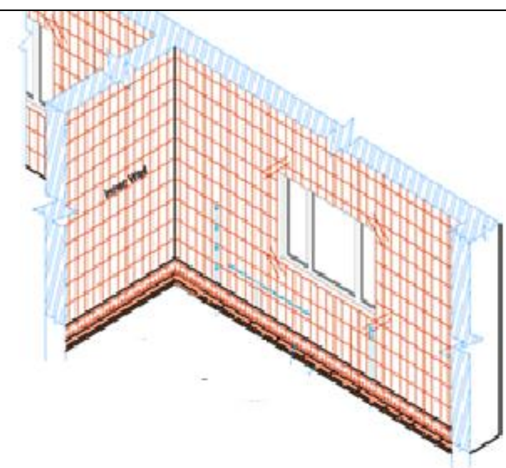
NSET is preparing the handbook on "Repair and Retrofit of Earthquake Damaged Stone in Mud Buildings" to facilitate the task of retrofitting the buildings damaged in Gorkha earthquake. This is only focused on stone in mud buildings, majority of damaged buildings in Gorkha earthquake 2015 which have been identified as damage grade 1 to 3 during damage assessment and are repairable and can be retrofitted but need not be demolished. The document will contain the damage grades in stone in mud buildings, identification of common repair methodologies for various levels of damage, retrofit methodologies and construction steps, drawings and design calculations of selected categories of buildings. Retrofit methods will include Splint and Bandage and jacketing using different locally available materials such as i) Steel bars ii) Welded wire mesh iii) Gabion wire iv) Timber v) Polypropylene band vi) Geo-Mesh etc. depending on accessibility of the materials. The location where steel and cement is available and transportation is easy RCC is suggested using steel or welded wire mesh and the sites where steel and cement is not available and relatively remote gabion wire, timber, PP-band or geo mesh is suggested depending on the availability of materials and ease of transportation (Photos and sketches included below). Design assumptions and philosophy, analysis and design methods, design details and drawings, cost estimation will also be included in the handbook to facilitate the engineer and implementer for concept and understanding. Three prototype designs are selected for retrofit design, one from Dhading and two from Nuwakot which are most commonly found in upper and lower Himalayan belts of Nepal. Though the handbook is targeted for stone in mud buildings that were damaged in 2015 Gorkha Earthquake, this document can also be used for any other stone in mud buildings that need repair and/or retrofit.



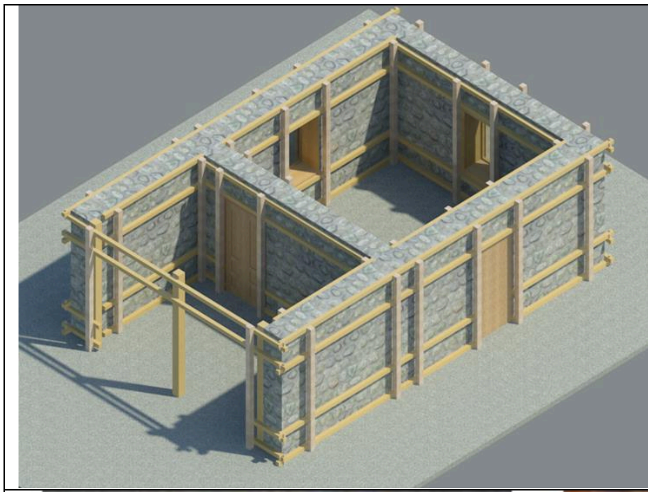
3D View of Splint and Bandage Method of Retrofitting



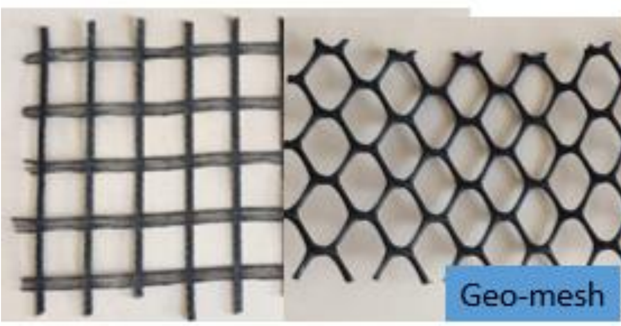
Schematic drawing of Splint and Bandage Method



Schematic drawing of Wall jacketing



Timber in Splint and Bandage Method of Retrofitting (Some rural areas of Nepal, timber is locally available and can be used for retrofiting. But other materials such as steel, cement, sand etc are not available and not easy to transport.)



Different kind of materials to be used for retrofitting depending on availability and ease of transport at the site

Gabion wire or PP-Band can be used to prevent the local failure of stone walls in between bands

<u>Research on Retrofit Techniques for Non-Engineered Housing - NSET</u>

The National Society for Earthquake Technology-Nepal (NSET) is working with the Beijing Normal University, China on a collaborative research project under the International Centre for Collaborative Research on Disaster Risk Reduction ICCR-DRR. The project is called “Development, testing, demonstration and training of better-built procedures and retrofit techniques for non-engineered housing in urban and rural areas of the Himalayan belt” and has financial support from the

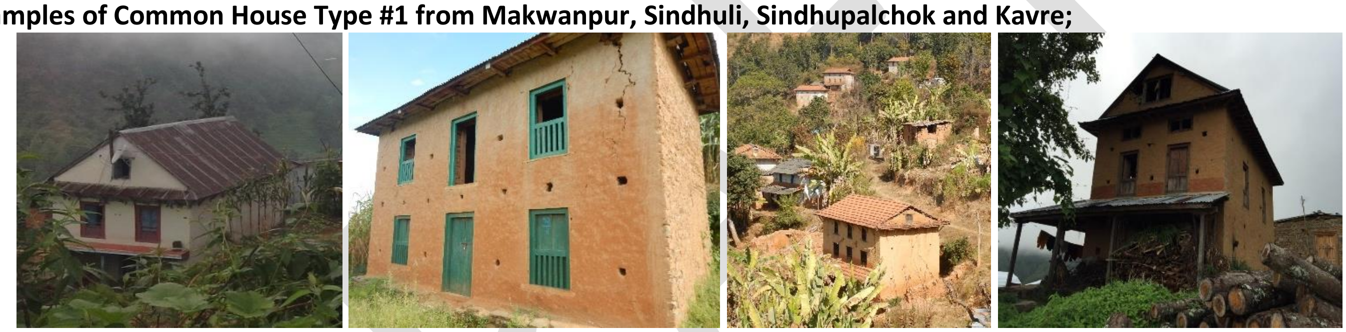
UK's Department for International Development (DFID). The research commenced on July 2016 and is scheduled to be complete in September 2017. The study is focused on low strength stone masonry buildings which suffered extensive damage in the Gorkha Earthquake 2015. The main purpose of the study is to do laboratory experimental tests on stone masonry buildings to understand the behaviour of the buildings in earthquakes and help develop a proper evidence based guideline for the construction and retrofit of such buildings. More details on the research are provided in a draft report, available here.

Type Design Approach – Build Change & NSET

Since there is neither the time nor the capacity to create individual, bespoke retrofit designs for every house, it is necessary to find ways streamline the retrofitting design and monitoring process: Essentially circumventing the need for individual, bespoke retrofit designs for the rural context by systematically providing tools and training which enable less experienced engineers to adapt and apply a centrally prepared retrofitting approach.

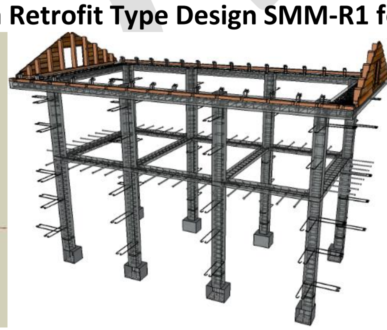
This can be done effectively within the reconstruction because of the cultural and technical homogeneity of Nepali rural houses. Indeed, in any given village, District or general geographic area, most rural houses follow similar patterns of configuration, size, materials and technology. As a result, houses can be sorted into typology sub-sets or house types which fall into simple specifications. For each housing type a standard retrofit methodology can be developed – a standard house type design.

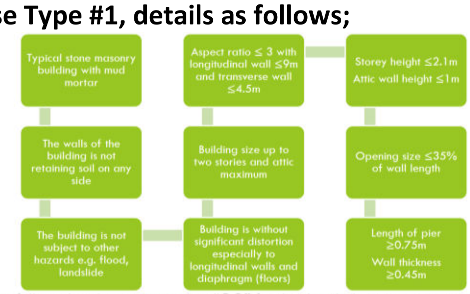
Examples of Common House Type #1 from Makwanpur, Sindhuli, Sindhupalchok and Kavre;



Through research and identification of the most common rural housing types in Nepal, a library of retrofitting house type designs can be developed tailor-made to each such type. This recommended technical method for a standardised and consistent implementation of retrofitting at scale in the Nepal earthquake reconstruction is through the development of standard house type retrofit designs and implementation packages. This approach would permit to streamline the retrofitting of 60% of the rural houses in need of such an intervention.

Build Change have developed a Retrofit Type Design SMM-R1 for House Type #1, details as follows;



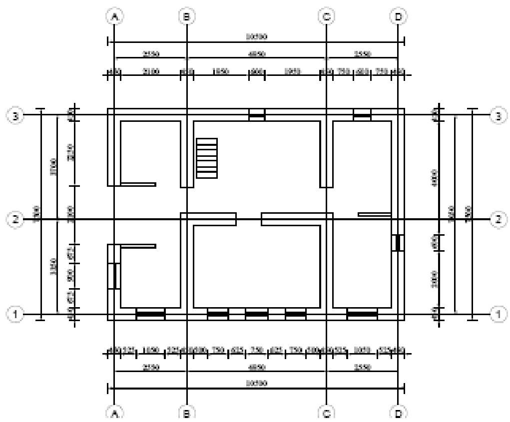


If the building follows the above criteria then "GO" for the detailed measurement and type design adaptation, if it does not then the "TYPE DESIGN" will not cover this house, hence "NO GO".

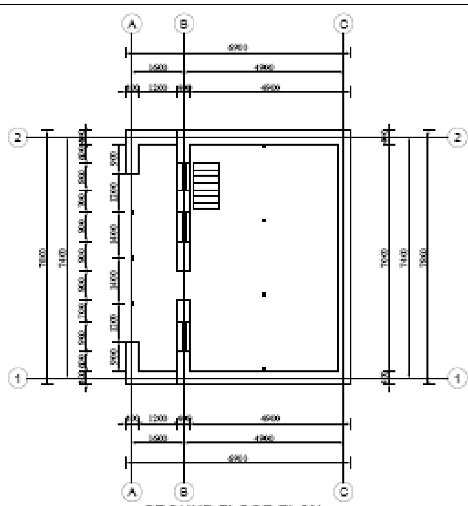
Sangachowk Example Retrofit - Before   - After

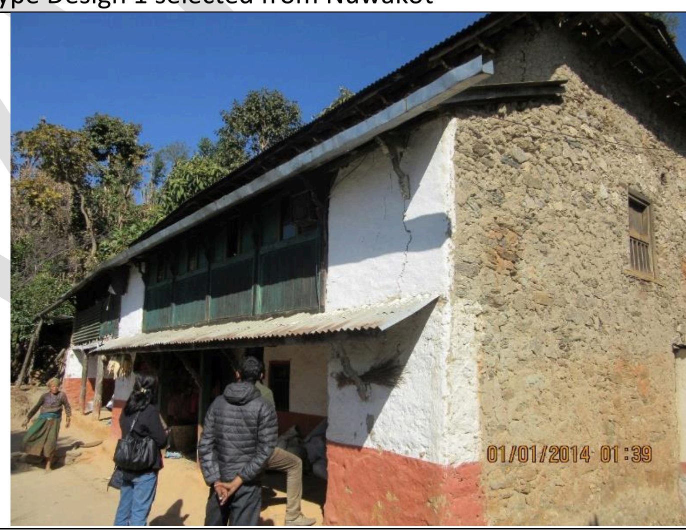



NSET are developing type design models based on the following typical structures:

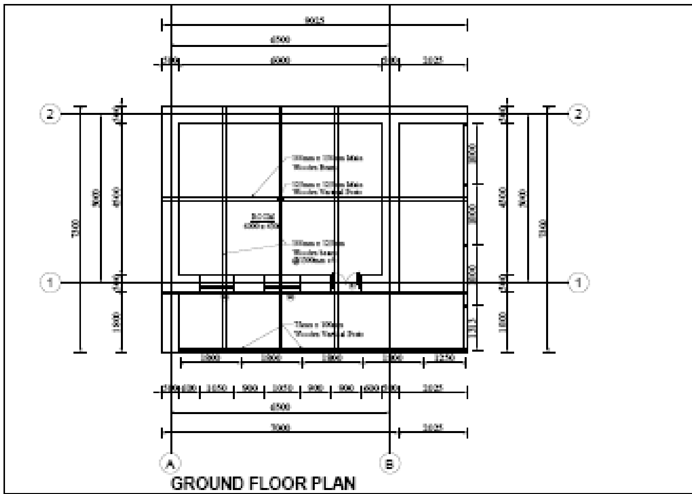



Plan and Building front view of Type Design 1 selected from Nuwakot


GROUND FLOOR PLAN



Plan and Building front view of Type Design 2 selected from Nuwakot



Plan and Building front view of Type Design 3 selected from Dhading

Summary Benefits of Retrofit Type Design Approach in Nepal:

- Large areas of traditional, vernacular architecture can be preserved and enhanced.
- Value for money for the homeowner; estimated <500 NPR/ft2 of living space.
- A robust, repeatable approach which seismically strengthens all sections of the house.
- Provides tools and training which enable less experienced engineers to systematically adapt and apply centrally prepared retrofitting approach based on sound engineering research.
- Reduces risks in retrofitting by standardising a practical adaptation, implementation and inspection approach and allows for rapid scale up.
- Enables various stakeholders including homeowners, partner organisations, private sector and social entrepreneurs to utilise the method based on sound engineering research.

There is a need to develop a type design library to include additional retrofitting type designs specifically for rural damaged houses as rebuilt and non-compliant houses in the northern VDCs. This should be a collaborative process, pooling technical resources of POs and GoN to maximise coverage of type designs developed.

Retrofitting Springboard Project – Build Change & UNOPS

This is a 6.5 month (28 week) long project being jointly implemented by Build Change and UNOPS. The project will be implemented in 18 locations through 3 cycles of approximately two months. Build Change has recommended that up to 6 districts be targeted, based on the concentration of different potential retrofitting types:

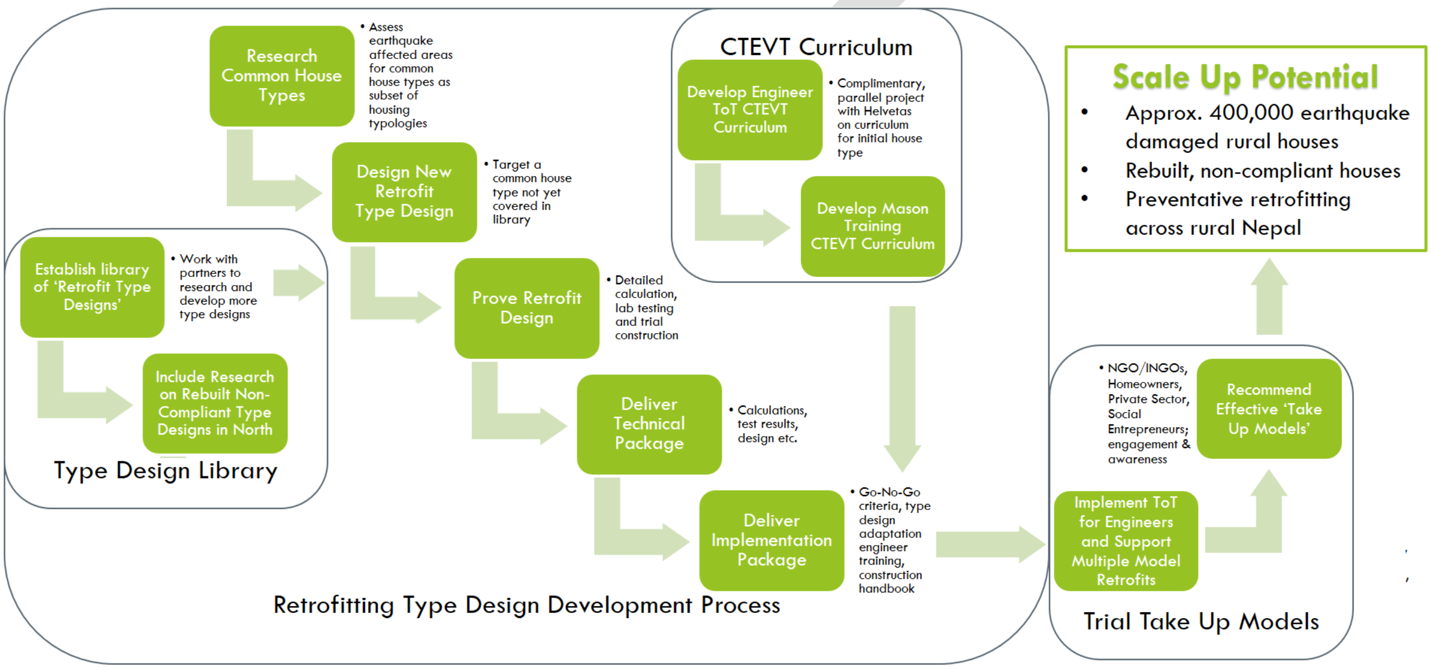
- **Houses partially damaged by the earthquake:** Example locations include Kavre, Sindhuli, Makwanpur (from 11 most affected rural districts) and Syanja (from 17 additional affected districts): application and adaptation of standard rural house type design #1. Development of additional type designs for common damaged house types where relevant.
- **Houses under construction or rebuilt since the earthquake but not meeting the NRA/DUDBC inspection criteria:** Example locations include Dhading, Gorkha, Sindhupalchowk, and Rasuwa: investigation in selected field locations to determine retrofit solutions for non-compliant new-builds and ongoing post-earthquake construction.
- Piloting and validation of 2 retrofit solutions, as well as a set of techniques for seismic upgrades of ongoing-construction and post-earthquake new-builds.

The quality monitoring will be carried out by Build Change and GoN engineers and it is expected that 100 engineers will be trained in total. This will include NRA / DUDBC / DL-PIU engineers, local engineers from private sector entrepreneurs, Partner Organisations (POs) engineering staff, and Build Change and UNOPS staff. In additions, between **100** and **280** masons from affected communities with high retrofit potential will also be trained, including 10 expert builder trainers.

The project will identify implementing partners as per the targeting above, and then locations for retrofits will be selected. A retrofit building assessment and damage analysis will be carried out along with related community awareness activities. Design adaptation based on the assessment and analysis will be completed and then on-site training and construction support will be provided. The project will include investigation in the field to determine retrofit solutions for new-builds and ongoing post-earthquake construction. The project will prepare a summary of best practices on engagement of communities, private sector entrepreneurs, POs, and Government of Nepal officials in retrofitting based on field experience and this will include technical advice on house type and Go-No Go criteria development for use in scale up.

In addition to the above, this project will practically test a range of retrofitting take-up models:

1. Owner driven model
2. Entrepreneur driven model – private sector / social entrepreneur
3. PO supported model – conversion of existing NRA approved ‘model houses’ to ‘model retrofits’ through partners



<u>Model Retrofit Trainings - Helvetas Employment Fund & Build Change</u>

The goal of this project is to contribute towards the progressive expansion of CTEVT's training curricula library, specifically in areas related to reduction of disaster-risk in construction and skills on retrofitting for engineers and local people in rural areas. The expected outputs of the project are as follows:

1. Within 7 months, CTEVT has in its possession the appropriately formatted and field tested written pedagogical program, assessment and material guides necessary to consider the formal adoption of a training course “curricula” on seismic retrofit evaluation and project management.
2. Retrofitting engineering ToT development; Selected local engineers from private sector and NGOs, NRA/DUDBC engineers, Helvetas Service Providers - Main Trainers etc. from Kavre and Sindhuli are trained and subsequently able to jointly, seismically evaluate and manage the selection of rural houses in Nepal, suitable for retrofitting.
3. Retrofitting builder training development; a small, selected number of vulnerable houses in Kavre and Sindhuli have been seismically evaluated and retrofitted delivering; retrofitted houses, builder training events, local builders trained.

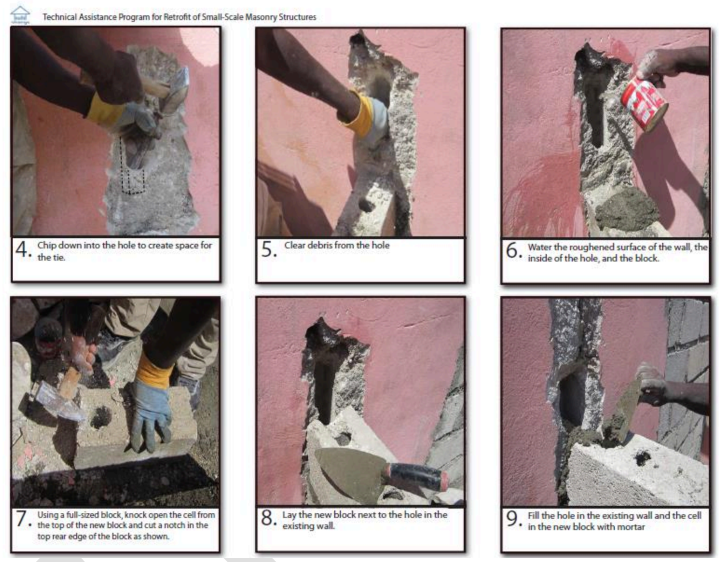


Figure 1 - Example; Build Change Retrofit Picture Guide; Visual Aid in the Execution of Seismic Retrofits, page 24

4. Retrofit-picture-guide, to be used as a training material for teaching engineers. Non-engineering retrofit –picture-guide for local people – the guide book focuses on non-structural damage or slight structural damage (Grade 1 and 2) where simple interventions can be done without engineering advice.